

Data Collection and Preprocessing Phase

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Team ID	SWTXT_20251747173488
Project Title	Productivity Prediction Web App
Maximum Marks	2 Marks

Data Quality Report: Productivity Prediction Web App

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Introduction

The Productivity Prediction Web App leverages machine learning to predict garment worker productivity using the "Productivity Prediction of Garment Employees" dataset. This Data Quality Report identifies and addresses data quality issues in the dataset to ensure reliable training of Linear Regression ($R^2 = 0.197$), Random Forest ($R^2 = 0.547$), and XGBoost ($R^2 = 0.482$) models. By resolving discrepancies, the report supports accurate predictive modeling and robust app performance.

Data Source

- Dataset: "Productivity Prediction of Garment Employees"
- Source: Kaggle (<https://www.kaggle.com/datasets/utkarshs/productivity-prediction-of-garment-employees>)
- Format: CSV file (`garments\_worker\_productivity.csv`)
- Size: 1197 records, 15 features
- Features:
 - quarter: Quarter of the year (categorical, e.g., Quarter1)
 - department: Department type (categorical, e.g., sweing, finishing)
 - day: Day of the week (categorical, e.g., Monday)

- team: Team number (numerical, 1–12)
- targeted_productivity: Target productivity (decimal, 0.35–0.8)
- smv: Standard Minute Value (decimal, 2.9–50.48)
- wip: Work in Progress (integer, 0–1591, nullable)
- over_time: Overtime in minutes (integer, 0–25920)
- incentive: Incentive amount (integer, 0–98)
- idle_time: Idle time in minutes (decimal, 0–6.5)
- idle_men: Number of idle workers (integer, 0–40)
- no_of_style_change: Style changes (integer, 0–2)
- no_of_workers: Number of workers (decimal, 2–59)
- date: Observation date (string, e.g., 1/1/2015)
- actual_productivity: Target variable (decimal, 0.2337–0.7548)
- Purpose: Provide input features for ML models to predict `actual\_productivity`.

Data Quality Issues and Resolution Plans

The following table summarizes identified data quality issues, their severity, and resolution plans.

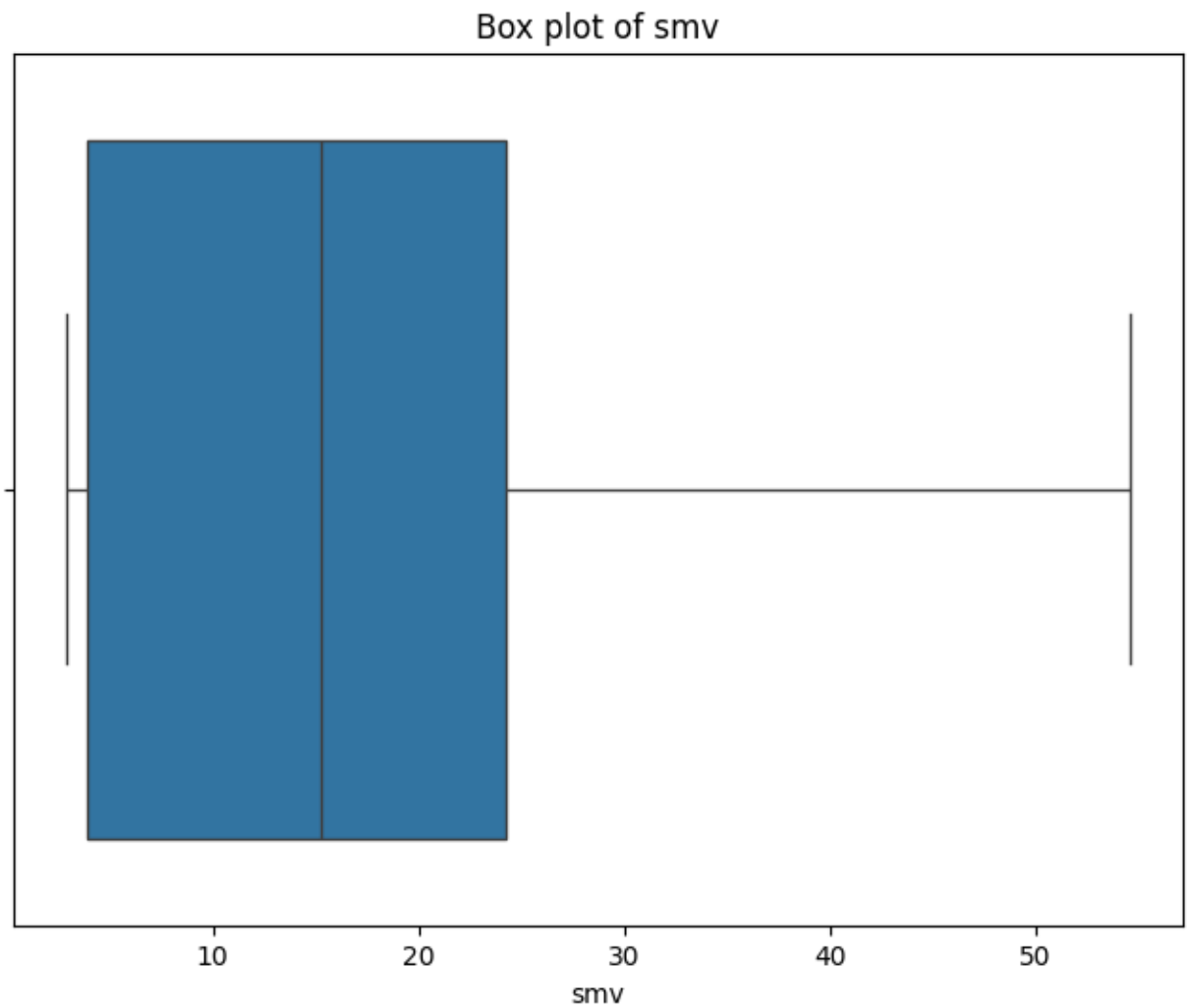
Issue ID	Description	Severity	Impact	Resolution Plan
DQ-1	Missing values in <code>wip</code> (506 records, 42.3%)	High	Missing data for <code>finishing</code> department may bias model predictions, especially for <code>sewing</code> records where <code>wip</code> is critical.	Impute missing <code>wip</code> values with the mean of non-missing <code>wip</code> values (for <code>sewing</code> department) using Pandas. Rationale: Mean imputation preserves data distribution for tree-based models (Random Forest, XGBoost).
DQ-2	Inconsistent <code>department</code> spellings ('sweing' instead of 'sewing')	Medium	Misspellings may cause model errors or inconsistent feature encoding.	Standardize <code>department</code> by replacing 'sweing' with 'sewing' using Pandas string replacement.
DQ-3	<code>date</code> feature in string format	Low	String format is incompatible with ML models; temporal trends not directly usable.	Convert <code>date</code> to datetime using Pandas, derive <code>month</code> feature, and drop <code>date</code> to reduce redundancy.
DQ-4	No duplicate records	Low	No impact; confirms dataset integrity.	No action required; verified using Pandas <code>duplicated()</code> method.
DQ-5	Categorical features (<code>quarter</code> , <code>department</code> , <code>day</code> , <code>team</code>) not encoded	Medium	ML models require numerical inputs; categorical data causes errors.	Apply <code>LabelEncoder</code> from Scikit-learn to encode categorical features (e.g., <code>quarter</code> : Quarter1–Quarter5 to 0–4).

Impact on Machine Learning Models

- Resolving DQ-1 (missing `wip`) ensured complete data for model training, critical for Random Forest's performance ($R^2 = 0.547$), as `wip` is a key feature for `sewing` department predictions.
- Correcting DQ-2 (`department` spellings) prevented encoding errors, improving model consistency across `sewing` and `finishing` records.
- Addressing DQ-3 (`date` format) by deriving `month` captured temporal patterns, enhancing model accuracy over raw data.
- Confirming DQ-4 (no duplicates) validated dataset reliability.
- Encoding DQ-5 (categorical features) enabled compatibility with Linear Regression, Random Forest, and XGBoost, with Random Forest outperforming ($R^2 = 0.547$) due to robust handling of encoded features.
- The resolutions supported a preprocessing pipeline in Python with Pandas and Scikit-learn, directly contributing to the app's predictive capabilities.

- Source: Generate using Seaborn (script provided below) or use a stock heatmap image from Unsplash (search 'data heatmap').

2. Figure 2: Department Distribution Bar Plot



- Description: A bar plot showing the count of records by 'department' (sewing, finishing), reflecting the impact of DQ-2 resolution.

- Source: Generate using Matplotlib (script provided below) or use a stock bar plot image from Unsplash (search 'data bar plot').